

Homework 2

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1. Analyze the Logical Forms

(a)

Let

J = John is telling the truth

and

B = Bill is telling the truth.

The logical form is

$$(J \wedge B) \vee (\neg J \wedge \neg B)$$

(b)

Let

F = I will have fish, C = I will have chicken, M = I will have mashed potatoes.

The logical form is

$$(F \vee C) \wedge \neg(F \wedge M)$$

(c)

The statement “3 is a common divisor of 6, 9, and 15” is written as

$$3 \mid 6 \wedge 3 \mid 9 \wedge 3 \mid 15$$

2. Well-Formed Formulas

(a)

$$\neg(\neg P \vee \neg\neg R)$$

Well-formed

(b)

$$\neg(P, Q, \wedge R)$$

Not well-formed

The commas are not logical connectives.

(c)

$$P \wedge \neg P$$

Well-formed

(d)

$$(P \wedge Q)(P \vee R)$$

Not well-formed

There is no logical connective between the two expressions.

3. Translate Formulas into English

Let

$$T = \text{Taxes will go up}$$

and

$$D = \text{The deficit will go up.}$$

(a)

$$T \vee D$$

Taxes will go up or the deficit will go up.

(b)

$$\neg(T \wedge D) \wedge \neg(\neg T \wedge \neg D)$$

Either taxes will go up or the deficit will go up, but not both.

(c)

$$(T \wedge \neg D) \vee (D \wedge \neg T)$$

Either taxes will go up and the deficit will not, or the deficit will go up and taxes will not.

4. Premises, Conclusions, and Validity

(a)

Let

J = Jane wins the math prize,

P = Pete wins the math prize,

and

C = Pete wins the chemistry prize.

Premises:

$$\neg(J \wedge P)$$

$$P \vee C$$

$$J$$

Conclusion:

$$C$$

Logical form:

$$\neg(J \wedge P), \quad P \vee C, \quad J \quad \therefore \quad C$$

The argument is **valid**.

Since Jane wins the math prize, Pete cannot also win the math prize. Therefore, Pete must win the chemistry prize.

(b)

Let

B = The main course is beef,

F = The main course is fish,

P = The vegetable is peas,

and

C = The vegetable is corn.

Premises:

$$B \vee F$$

$$P \vee C$$

$$\neg(F \wedge C)$$

Conclusion:

$$\neg(B \wedge P)$$

Logical form:

$$B \vee F, \quad P \vee C, \quad \neg(F \wedge C) \quad \therefore \quad \neg(B \wedge P)$$

The argument is **invalid**.

For example, beef and peas satisfies all three premises while making the conclusion false.

(c)

Let

J = John is telling the truth,

B = Bill is telling the truth,

and

S = Sam is telling the truth.

Premises:

$$J \vee B$$

$$\neg S \vee \neg B$$

Conclusion:

$$J \vee \neg S$$

Logical form:

$$J \vee B, \quad \neg S \vee \neg B \quad \therefore \quad J \vee \neg S$$

The argument is **valid**.

5. Truth Tables

(a)

$$\neg[P \wedge (Q \vee \neg P)]$$

P	Q	$\neg P$	$Q \vee \neg P$	$P \wedge (Q \vee \neg P)$	$\neg[P \wedge (Q \vee \neg P)]$
T	T	F	T	T	F
T	F	F	F	F	T
F	T	T	T	F	T
F	F	T	T	F	T

(b)

$$(P \vee Q) \wedge (\neg P \vee R)$$

P	Q	R	$P \vee Q$	$\neg P$	$\neg P \vee R$	$(P \vee Q) \wedge (\neg P \vee R)$
T	T	T	T	F	T	T
T	T	F	T	F	F	F
T	F	T	T	F	T	T
T	F	F	T	F	F	F
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	F
F	F	F	F	T	T	F

6. Exclusive OR

The symbol $+$ means exclusive OR:

$$P + Q = P \text{ or } Q, \text{ but not both.}$$

(a)

P	Q	$P + Q$
T	T	F
T	F	T
F	T	T
F	F	F

(b)

An equivalent formula using only $\wedge$, $\vee$, and $\neg$ is

$$(P \vee Q) \wedge \neg(P \wedge Q)$$

Truth table:

P	Q	$P \vee Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$(P \vee Q) \wedge \neg(P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

The final column is identical to the truth table for $P + Q$.

Therefore,

$$P + Q \equiv (P \vee Q) \wedge \neg(P \wedge Q)$$

7. Formula Using Only $\wedge$ and $\neg$

Find a formula equivalent to $P \vee Q$ using only $\wedge$ and $\neg$.

By De Morgan's Law,

$$P \vee Q \equiv \neg(\neg P \wedge \neg Q)$$

Truth table:

P	Q	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$\neg(\neg P \wedge \neg Q)$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	F

The final column is identical to $P \vee Q$.

8. NAND

The symbol $|$ means NAND:

$$P|Q = \neg(P \wedge Q)$$

(a)

P	Q	$P \wedge Q$	$P Q$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

(b)

$$P|Q \equiv \neg(P \wedge Q)$$

(c)

Using only the connective $|$:

For $\neg P$:

$$\neg P \equiv P|P$$

For $P \vee Q$:

$$P \vee Q \equiv (P|P)|(Q|Q)$$

For $P \wedge Q$:

$$P \wedge Q \equiv (P|Q)|(P|Q)$$

9. Formula Equivalence

The formulas are:

- (a) $(P \wedge Q) \vee (\neg P \wedge \neg Q)$
- (b) $\neg P \vee Q$
- (c) $(P \vee \neg Q) \wedge (Q \vee \neg P)$
- (d) $\neg(P \vee Q)$
- (e) $(Q \wedge P) \vee \neg P$

Truth table:

P	Q	(a)	(b)	(c)	(d)	(e)
T	T	T	T	T	F	T
T	F	F	F	F	F	F
F	T	F	T	F	F	T
F	F	T	T	T	T	T

Comparing the final columns:

$$(a) \equiv (c)$$

and

$$(b) \equiv (e)$$

Formula (d) is not equivalent to any of the other formulas.